

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET D — ASSERTION & REASONING SPECIAL

CHAPTER 06: HOW FORCES AFFECT MOTION & LAWS OF MOTION

TIME ALLOWED: 45 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-D-06

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET D - ASSERTION & REASONING SPECIAL):

1. The question paper comprises **25 Assertion-Reasoning Questions** of **1 mark each**.
2. Each question consists of two statements — **Assertion (A)** and **Reason (R)**.
3. Select the correct option from the following 4 standardized choices:
 - **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 - **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 - **(C)** Assertion (A) is true but Reason (R) is false.
 - **(D)** Assertion (A) is false but Reason (R) is true.
4. **Section A (Q1 – Q10):** Core Theoretical Principles & Scientific Definitions.
5. **Section B (Q11 – Q18):** Experimental Observations & Phenomenological Inferences.
6. **Section C (Q19 – Q25):** Advanced Conceptual Nuances & Common Scientific Misconceptions.
7. All questions are compulsory. There is no negative marking.
8. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

STANDARD DIRECTIONS FOR ALL 25 ASSERTION & REASONING QUESTIONS:

- (A)** Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

SECTION A: CORE THEORETICAL PRINCIPLES (Q.1 TO Q.10)

[10 MARKS]

Q.1

Assertion (A): Balanced forces acting on a body do not alter its state of rest or of uniform motion in a straight line.

Reason (R): The vector sum (resultant) of all balanced forces acting on a body is exactly equal to zero ($\sum F_{\text{net}} = 0$).

- | | |
|---|---|
| <p>(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).</p> <p>(C) Assertion (A) is true but Reason (R) is false.</p> | <p>(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).</p> <p>(D) Assertion (A) is false but Reason (R) is true.</p> |
|---|---|

Q.2

Assertion (A): A heavier iron anvil of mass 50 kg has greater inertia than a wooden block of mass 5 kg.

Reason (R): The mass of an object is a quantitative measure of its inertia.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.3

Assertion (A): According to Newton's first law of motion, force is required to maintain a moving object at constant velocity in deep space.

Reason (R): A body continues in its state of rest or uniform straight-line motion unless acted upon by an external net unbalanced force.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.4

Assertion (A): The rate of change of linear momentum of a body is directly proportional to the applied unbalanced force and takes place in the direction of the force.

Reason (R): Newton's second law of motion provides a quantitative mathematical relationship for calculating force ($F = ma$).

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.5

Assertion (A): Action and reaction forces never cancel each other to produce equilibrium on a single body.

Reason (R): Action and reaction forces always act simultaneously on two different, distinct physical bodies.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.6

Assertion (A): The SI unit of linear momentum is kilogram meter per second ($\text{kg} \cdot \text{m/s}$), which is equivalent to Newton second ($\text{N} \cdot \text{s}$).

Reason (R): Momentum is defined as the product of mass (m) and velocity (v), having dimensions $[\text{M}][\text{L}][\text{T}]^{-1}$.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.7

Assertion (A): A high-velocity bullet of mass 10 g can easily penetrate a thick wooden plank, whereas an identical bullet thrown gently by hand bounces off.

Reason (R): The penetrating power and impact damage of an object depend on its linear momentum ($p = mv$), not on mass alone.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.8

Assertion (A): Friction between the soles of shoes and the ground is an essential force that enables human walking.

Reason (R): When walking, a person pushes backward on the ground; according to Newton's third law, the ground pushes forward with equal force.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.9

Assertion (A): A body moving with constant speed along a circular track has constant momentum.

Reason (R): Linear momentum is a vector quantity having the same direction as the velocity vector, which changes continuously in circular motion.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.10

Assertion (A): Newton's first law of motion is also known as the Law of Inertia.

Reason (R): Galileo Galilei first deduced that objects in motion continue moving with constant velocity if no resistive forces act on them.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

SECTION B: EXPERIMENTAL OBSERVATIONS & INFERENCES (Q.11 TO Q.18)

[8 MARKS]

Q.11

Assertion (A): A cricket fielder pulls his hands backwards while catching a fast-flying ball.

Reason (R): Pulling hands back increases the duration of time over which momentum drops to zero, thereby reducing the impact force on his palms.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.12

Assertion (A): When a carpet is beaten vigorously with a stick, dust particles fly out.

Reason (R): Due to inertia of rest, dust particles tend to stay in their stationary positions while the carpet fabric is suddenly jerked away.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.13

Assertion (A): Passengers standing in a moving bus fall forward when the driver suddenly applies emergency brakes.

Reason (R): The lower part of the body in contact with the bus floor comes to rest abruptly, while the upper torso continues moving forward due to inertia of motion.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.14

Assertion (A): When a gun fires a bullet forward, the gun recoils backwards with a lower velocity than the bullet.

Reason (R): According to the law of conservation of momentum, the backward recoil momentum of the gun equals the forward momentum of the bullet; since the gun has a much larger mass, its recoil velocity is proportionally smaller.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.15

Assertion (A): Modern passenger cars are equipped with crumple zones and seatbelts.

Reason (R): Crumple zones prolong collision impact time, drastically reducing the deceleration and peak force transmitted to passengers.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.16

Assertion (A): Athletes in high jump events are made to land on deep foam mattresses or soft sand pits.

Reason (R): A soft landing increases the time taken for the athlete's vertical momentum to drop to zero, reducing the ground reaction force.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.17

Assertion (A): A rocket operates on the principle of Newton's third law of motion and linear momentum conservation.

Reason (R): A rocket pushes against the atmospheric air surrounding it to propel itself into outer space.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.18

Assertion (A): When a rotating stone tied to a string is released, it flies off tangentially in a straight line.

Reason (R): The moment the centripetal inward tension force ceases, the stone continues in the direction of its instantaneous tangential velocity due to inertia.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

SECTION C: ADVANCED NUANCES & MISCONCEPTIONS (Q.19 TO Q.25)

[7 MARKS]

Q.19

Assertion (A): In any isolated system of colliding bodies where no external net force acts, total linear momentum is conserved.

Reason (R): The internal collision forces between interacting bodies occur in equal and opposite action-reaction pairs according to Newton's third law.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.20

Assertion (A): A horse must exert a continuous forward pull to keep a heavy cart moving with uniform constant velocity on a rough dirt road.

Reason (R): The continuous forward pull of the horse is required to overcome the opposing resistive friction force exerted by the ground.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.21

Assertion (A): An astronaut of mass 80 kg floating freely outside the International Space Station throws a 2 kg metal wrench forward at 10 m/s; the astronaut drifts backwards.

Reason (R): The backward drift velocity of the astronaut is 0.25 m/s in accordance with the law of conservation of momentum.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.22

Assertion (A): Water droplets fly off radially when an umbrella is rotated rapidly.

Reason (R): The adhesive force between water and the umbrella fabric is insufficient to provide the necessary centripetal force for circular motion.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.23

Assertion (A): If the net force acting on a moving particle is zero, its kinetic energy must remain constant.

Reason (R): When net force is zero, acceleration is zero, meaning velocity (and hence speed) remains strictly constant.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.24

Assertion (A): A heavy truck and a lightweight bicycle moving with identical kinetic energy have identical linear momentum.

Reason (R): Linear momentum and kinetic energy are related by the formula $p = \sqrt{2m E_k}$; for equal E_k , momentum is directly proportional to $\sqrt{m}$.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.25

Assertion (A): A swimmer pushes the water backwards with his hands and feet to propel himself forward through the pool.

Reason (R): According to Newton's third law of motion, for every action force there is an equal and opposite reaction force.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET D)

01	☐ A ☐ B ☐ C ☐ D	02	☐ A ☐ B ☐ C ☐ D	03	☐ A ☐ B ☐ C ☐ D	04	☐ A ☐ B ☐ C ☐ D	05	☐ A ☐ B ☐ C ☐ D
06	☐ A ☐ B ☐ C ☐ D	07	☐ A ☐ B ☐ C ☐ D	08	☐ A ☐ B ☐ C ☐ D	09	☐ A ☐ B ☐ C ☐ D	10	☐ A ☐ B ☐ C ☐ D
11	☐ A ☐ B ☐ C ☐ D	12	☐ A ☐ B ☐ C ☐ D	13	☐ A ☐ B ☐ C ☐ D	14	☐ A ☐ B ☐ C ☐ D	15	☐ A ☐ B ☐ C ☐ D
16	☐ A ☐ B ☐ C ☐ D	17	☐ A ☐ B ☐ C ☐ D	18	☐ A ☐ B ☐ C ☐ D	19	☐ A ☐ B ☐ C ☐ D	20	☐ A ☐ B ☐ C ☐ D
21	☐ A ☐ B ☐ C ☐ D	22	☐ A ☐ B ☐ C ☐ D	23	☐ A ☐ B ☐ C ☐ D	24	☐ A ☐ B ☐ C ☐ D	25	☐ A ☐ B ☐ C ☐ D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & SCIENTIFIC CAUSAL EXPLANATIONS (SET D)

Q. No.	Correct	Scientific Rationale & Causal Analysis
Q.1	(A)	Assertion is true. Reason is true and directly explains why balanced forces cause zero linear acceleration.
Q.2	(A)	Assertion is true. Reason is true and states Newton's fundamental definition: mass is the measure of inertia.
Q.3	(D)	Assertion is false: Zero force is needed to maintain constant velocity in frictionless space. Reason is true (Newton's 1st Law).
Q.4	(A)	Assertion is true. Reason is true and connects the rate of change of momentum directly to Newton's second law.
Q.5	(A)	Assertion is true. Reason is true and clarifies the crucial distinction in Newton's 3rd Law preventing mutual cancellation.
Q.6	(A)	Assertion is true. Reason is true and explains the dimensional and algebraic derivation of the momentum unit.
Q.7	(A)	Assertion is true. Reason is true and explains why high velocity creates massive momentum despite small mass.
Q.8	(A)	Assertion is true. Reason is true and explains the action-reaction mechanism with ground friction during walking.
Q.9	(D)	Assertion is false: Direction of velocity continuously changes, so momentum changes continuously. Reason is true.
Q.10	(B)	Both statements are true historical facts, but Reason cites Galileo's contribution rather than defining why Newton's first law is called the law of inertia.
Q.11	(A)	Assertion is true. Reason is true and applies $F = \frac{\Delta p}{\Delta t}$; increasing Δt drastically lowers the impact force F .
Q.12	(A)	Assertion is true. Reason is true and correctly attributes the separation of dust to inertia of rest.
Q.13	(A)	Assertion is true. Reason is true and provides the physiological and physical explanation of inertia of motion.
Q.14	(A)	Assertion is true. Reason is true and explains the inverse relationship between recoil velocity and mass ($v_{\text{gun}} = -\frac{m_{\text{bullet}} v_{\text{bullet}}}{M_{\text{gun}}}$).
Q.15	(A)	Assertion is true. Reason is true and explains the life-saving physics of impulse and force attenuation.
Q.16	(A)	Assertion is true. Reason is true and applies Newton's second law ($F = \Delta p / \Delta t$) to prevent injury.
Q.17	(C)	Assertion is true. Reason is false: Rockets do not push against atmospheric air; they propel by ejecting high-velocity exhaust gases backwards ($F = v_{\text{ex}} \frac{dm}{dt}$), working perfectly in a vacuum.
Q.18	(A)	Assertion is true. Reason is true and explains why removing centripetal constraint allows straight-line tangential inertial flight.
Q.19	(A)	Assertion is true. Reason is true and proves how Newton's third law directly leads to the law of conservation of momentum.
Q.20	(A)	Assertion is true. Reason is true and reconciles Aristotelian everyday observations with Newtonian physics.
Q.21	(A)	Assertion is true. Reason is true and correctly calculates: $80 \times v = 2 \times 10 \rightarrow v = 20/80 = 0.25 \text{ m/s}$.
Q.22	(A)	Assertion is true. Reason is true and explains droplet detachment when required centripetal force exceeds adhesion.
Q.23	(A)	Assertion is true. Reason is true and connects zero net force to constant speed and invariant kinetic energy.
Q.24	(D)	Assertion is false: The heavier truck has much greater momentum ($p = \sqrt{2m E_k}$). Reason is true.
Q.25	(A)	Assertion is true. Reason is true and correctly cites Newton's third law as the propulsion mechanism in swimming.